

Roudy Hajo

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Final-year Computer Science Engineering student specializing in Networks, Cloud Computing, and Distributed Systems at Polytech Nantes. Seeking a 22-week Software Engineering / Site Reliability Engineering (SWE/SRE) internship. Available from March 2027 through August 2027 (covering the full Summer 2027 program). Expected Graduation: August 2027.

EDUCATION

Polytech Nantes School of Engineering <i>Master's Degree in Computer Science (Networks, Security, Cloud, Distributed Systems)</i>	Nantes, France Sep. 2024 – Aug. 2027
University Institute of Technology of Nantes <i>Bachelor of Science in Computer Science (Data Structures, Algorithms, OOP, DBMS)</i>	Nantes, France Sep. 2022 – Jun. 2024

EXPERIENCE

Research Intern – Distributed Systems & DBMS <i>Coimbra Institute of Engineering</i>	May 2026 – Aug. 2026 Coimbra, Portugal
- Engineered a multi-agent QNN and reinforcement learning framework (Python/Qiskit) for automated index tuning, paired with an adaptive router balancing query loads across distributed replicas. - Reduced TPC-H benchmark execution time by ~50% under high concurrency while mitigating node skew; co-authoring a research paper on the findings (in progress).	
Software Engineer Intern <i>Nantes University – IT and Digital Services Department</i>	May 2025 – July 2025 Nantes, France
- Migrated the university course catalog platform 'Syllabus' to modern PHP and Symfony releases. - Streamlined CI/CD deployment pipelines and refactored core backend services to improve system reliability.	
Software Engineer Intern <i>Kookline – Web Development Agency</i>	Apr. 2024 – June 2024 Nantes, France
- Engineered a custom e-commerce and booking platform in PHP/MySQL for a skydiving service, processing reservation workflows and payments. - Developed custom modular backend extensions and hooks to integrate dynamic content and automated editorial publishing.	
Academic Tutor in Computer Science <i>University Institute of Technology of Nantes</i>	Oct. 2023 – July 2024 Nantes, France
Guided undergraduate students through advanced programming, core data structures, graph algorithms, and OOP design principles.	

PROJECTS

I2PoFM (Covert Internet over Radio) <i>Python, GNU Radio, I2P, SDR, Linux</i>	Feb. 2026 – Mar. 2026
- Engineered a physical-layer protocol multiplexing encrypted IP packets over FM broadcasts using DSSS and PCM. - Deployed an autonomous routing gateway on Raspberry Pi nodes interfaced with I2P for fault-tolerant packet delivery and network security.	
SketchOscent <i>Python, JavaScript, Webpack, Cheminformatics</i>	Sep. 2025 – May 2026
- Built algorithmic mapping routines connecting chemical functional groups to target olfactory descriptors. - Configured an automated Webpack AST obfuscation pipeline to secure proprietary processing code.	
PolyRisk – Autonomous Game Engine <i>Python, Algorithmic Decision Systems</i>	Oct. 2025 – Nov. 2025
- Implemented Monte Carlo Tree Search (MCTS) agents with custom heuristics for complex state-space evaluation. - Optimized tree traversal pruning to accelerate simulated decision rollouts across game branches.	
Dice Wars – Strategy Game Engine <i>C++, Object-Oriented Design</i>	Nov. 2025 – Jan. 2026
- Built a turn-based strategy game featuring procedural map generation via Voronoi partitioning and graph connectivity. - Designed deterministic AI opponents using minimax-based evaluation while enforcing strict memory bounds.	
Particles Engine <i>Go, Ebiten, Concurrent Computing, Data Structures</i>	Dec. 2022 – Jan. 2023
- Implemented a 2D physics engine using Go goroutines and worker pools for concurrent numerical simulation. - Optimized memory allocation and reduced GC pauses to maintain stable 60 FPS real-time rendering.	

TECHNICAL SKILLS

Languages: Go, C/C++, Python, Java, Rust, Kotlin, SQL (PostgreSQL, MariaDB), JavaScript, Bash
Systems & Cloud Infrastructure: Linux/Unix, Docker, Kubernetes, Google Cloud, AWS, Git, CI/CD, TCP/IP, DNS
Frameworks & Libraries: Spring Boot, React, Node.js, Symfony, Flask, JUnit, GNU Radio, gRPC
Spoken Languages: French (Native), English (Professional B2), Arabic (Fluent), Kurdish (Native)